



King's Research Portal

DOI:

[10.1016/j.chemosphere.2019.03.173](https://doi.org/10.1016/j.chemosphere.2019.03.173)

Document Version

Peer reviewed version

[Link to publication record in King's Research Portal](#)

Citation for published version (APA):

Mason, T. G., Chan, K. P., Schooling, C. M., Sun, S., Yang, A., Barratt, B., & Tian, L. (2019). Air quality changes after Hong Kong shipping emission policy: An accountability study. *Chemosphere*, 226, 616-624. <https://doi.org/10.1016/j.chemosphere.2019.03.173>

Citing this paper

Please note that where the full-text provided on King's Research Portal is the Author Accepted Manuscript or Post-Print version this may differ from the final Published version. If citing, it is advised that you check and use the publisher's definitive version for pagination, volume/issue, and date of publication details. And where the final published version is provided on the Research Portal, if citing you are again advised to check the publisher's website for any subsequent corrections.

General rights

Copyright and moral rights for the publications made accessible in the Research Portal are retained by the authors and/or other copyright owners and it is a condition of accessing publications that users recognize and abide by the legal requirements associated with these rights.

- Users may download and print one copy of any publication from the Research Portal for the purpose of private study or research.
- You may not further distribute the material or use it for any profit-making activity or commercial gain
- You may freely distribute the URL identifying the publication in the Research Portal

Take down policy

If you believe that this document breaches copyright please contact librarypure@kcl.ac.uk providing details, and we will remove access to the work immediately and investigate your claim.

**Air Quality changes after Hong Kong shipping emission policy:
An accountability study**

Tonya G. Mason^a, King Pan Chan^a, C. Mary Schooling^a, Shengzhi Sun^c, Aimin Yang^a, Yang
Yang^a, Benjamin Barratt^b, Linwei Tian^a

^a School of Public Health, Li Ka Shing Faculty of Medicine, The University of Hong Kong,
SAR, China;

^b MRC-PHE Centre for Environment and Health & NIHR GSTFT/KCL Biomedical Research
Centre, Analytical and Environmental Sciences Division, Faculty of Life Sciences & Medicine,
King's College London, London, UK.

^c Department of Epidemiology, Brown University School of Public Health, Providence, RI, USA

*Corresponding author:

Dr. Linwei Tian, School of Public Health, Li Ka Shing Faculty of Medicine, The University of
Hong Kong, Patrick Manson Building North Wing, 7 Sassoon Road, Pokfulam, Hong Kong
SAR, China. Phone (+852) 3917 9280; Email: linweit@hku.hk

Abstract

Background: On July 1st 2015, Hong Kong became the first city in Asia to implement a policy regulating sulfur dioxide (SO₂) in shipping emissions. We conducted an accountability study assessing the improvement in ambient air quality and estimating the effect on health outcomes of the policy.

Method: We used interrupted time series (ITS) with segmented regression to identify any change in ambient concentrations of SO₂ in contrast to other ambient pollutants (particulate matter <10 micrometers in diameter (PM₁₀), nitrogen dioxide (NO₂) and ozone (O₃)) at 10 monitoring stations in Hong Kong from 2010-2017. We validated these findings using cumulative sum control (CUSUM) charts. We used a validated risk assessment model to estimate effects of changes in air quality on death for natural causes, cardiovascular and respiratory diseases.

Results: Mean monthly concentrations of SO₂ fell abruptly at the monitoring station closest to the main shipping port (Kwai Chung (KC)) by -10.0µgm³ p-value= 0.0004), but not elsewhere. No such changes were evident for the other pollutants (PM₁₀, NO₂, O₃). CUSUM charts confirmed a change in July 2015. Estimated deaths avoided per year as a result of the policy were 379, 72, 30 for all natural causes, respiratory and cardiovascular diseases respectively.

Conclusion: Implementation of the shipping emission policy in Hong Kong successfully reduced ambient SO₂, with the potential to reduce mortality. However, to gain full benefits, restrictions on shipping emissions need to be implemented throughout the region.

Keywords: Shipping emission policy; Interrupted time series; segmented regression analysis; CUSUM; Sulfur dioxide; Kwai Chung.

1. Introduction

Globally, shipping emissions contribute significantly to air pollution, especially in areas situated around the coastline. About 70% of shipping emissions occur within 400 kilometers of land (Endresen et al. 2003)(Viana, M.; Hammingh, P.; Colette, A.; Querol 2014) exposing populations to anthropogenic emissions of particular matter (PM) and of sulfur and nitrogen oxides(Eyring et al. 2005) (Corbett, J J, Winerbrake, J.J., Green, E.H. 2007). Shipping emissions account for approximately 15% of global anthropogenic nitrogen oxides and 5-8% of global sulfur oxides (Eyring et al. 2005)(Corbett, J J, Winerbrake, J.J., Green, E.H. 2007). SO₂ has been known to cause cardiorespiratory health effects due to its ability to be a respiratory irritant and a bronchoconstrictor—resulting in cardiovascular abnormalities (Kan et al. 2010). Many studies have linked air pollution from SO₂ to adverse health effects for both morbidity and mortality for cardiorespiratory diseases (Katsouyanni et al. 1997b; Stieb et al. 2002, 2003) , including a 4 city study (Hong Kong, Shanghai, Wuhan and Bangkok) done in Asia—in 2010 (Kan et al. 2010) . Throughout time, sulfur rich fuels have proven to affect morbidity and mortality rates for cardiovascular and respiratory diseases however reduction in air pollution from these sulfur-rich fuels have resulted in lower cardiorespiratory deaths and hospital admissions (Dockery et al. 2013; Hedley et al. 2002; Stallings-Smith et al. 2013; Wong et al. 1998).

Hong Kong has the fourth busiest shipping port in the world and serves as a major hub port for South Asian Pacific region and the Mainland of China (Ng et al. 2013). Ship exhaust is one of the major sources of sulfur dioxide (SO₂) emissions in Hong Kong, contributing about 36% of ambient SO₂ concentrations, measured by monitors located close major shipping ports (Kwai Chung and Tsing Yi) (Yau et al. 2012). Lai et al. recently showed an annual excess death rate

due to marine emissions in Hong Kong, to be 9-20 times higher than those in the Pearl River Delta (PRD) regions (Lai et al. 2013). In 2012 the civic exchange department in Hong Kong, reported : SO₂ emissions from ocean going vessels was responsible for 519 premature deaths per year in the Pearl River Delta region, with majority of these deaths occurring in Hong Kong (385 avoidable deaths) (Kilburn et al. 2012). This is an indication that due to the heavy shipping traffic and a shipping route within close proximity to densely populated areas, in Hong Kong, marine pollution has become of major public health concern. Studies from the current decade have approximated about 3.8 million people living close marine ports in Hong Kong are directly exposed to shipping emissions that are high in SO₂ and other shipping related pollutants such as NO_x and PM₁₀ (Kilburn et al. 2012; Lai et al. 2013; Ng 2013).

Regulatory frameworks have been introduced to reduce the sulfur content of marine fuel internationally (International Maritime Organization 2015) and regionally (North-West Ports Clean Air Strategy) (San Pedro Bay Ports Clean Air Action Plan.). Implementation of such policies improve air quality and has the potential to reduce cardiopulmonary and lung cancer related mortality (Velders et al. 2011)(Winebrake et al. 2009). In a global city with the fourth busiest port in the world, we took advantage of the introduction of a shipping emission policy, implemented on July 1st 2015, requiring all ocean going vessels (OGV) to switch to fuel with a sulfur content not exceeding 0.5% at berth (Environmental Protection Department 2015), to assess the effects on air quality and to estimate the effects on mortality. In this study, we used interrupted time series (ITS) to quantify changes in SO₂ concentration and cumulative sum (CUSUM) charts to confirm that the timing of the changes coincided with the policy implementation in Hong Kong. We similarly assessed the changes in PM₁₀, O₃ and NO₂ to assess

the specificity of any changes and the changes by proximity to the port. We also estimated effects on deaths based on the excess risk associated with a $10\mu\text{gm}^3$ change in air pollutants.

2. Method

2.2. Locations

Figure 1 displays a map of Hong Kong with the 10 monitoring stations including the major shipping port, Kwai Chung (KC), where most container vessels berth. KC is also a hot spot for SO_2 emissions from container vessels (Ng et al. 2013). Monitoring stations at Tai Po (TP), Yuen Long (YL) and Tap Mun (TPM) are furthest from the port, and were considered as control stations.

2.1. Air pollutants

SO_2 was the main pollutant considered. PM_{10} , O_3 and NO_2 were considered as control outcomes because they would not be expected to change significantly as a result of the shipping emission policy on reducing the sulfur content in fuel. Daily concentrations of SO_2 , PM_{10} , NO_2 and O_3 were obtained from 10 of the 16 monitoring stations in Hong Kong, from January 1st, 2010 to December 31st, 2017. 3 roadside stations and 3 general stations with extensive missing data (Tseung Kwan O, Tuen Mun, and Kwun Tong) were excluded from the study. Exposure to shipping air pollution is better represented by general monitoring stations than road side stations (Huang et al. 2017). Most of Hong Kong's population lives within 5 kilometers of the 10 general monitoring stations in this study (Huang et al. 2017). All stations are operated by the Hong Kong Environmental Protection Department (HKEPD) who uses automatic analyzers at each monitoring station to obtain hourly ambient air pollution concentration readings.

108

109 *2.3. Air quality assessment*

110 Two complimentary methods were used to assess the impact of the implementation of the
111 shipping emission policy on air quality overall and by monitoring station. The first method, ITS
112 is a study design, commonly used for evaluating interventions (Dennis et al. 2013) . It is known
113 as one of the strongest quasi-experimental designs because the design automatically controls for
114 base line and secular trends (Dennis et al. 2013). Segmented regression is the statistical method
115 used for ITS, which allows us to visually and statistically assess air quality before (January 2013-
116 June 2015) and after (July 2015-December 2017) the implementation of the policy, in terms of
117 an abrupt and gradual changes(Nistal-Nuño 2017) (for more information see supplementary
118 appendix A).

119 The shipping policy was represented in the model below using an indicator variable(β_2) which
120 was assigned the value of 0 for the pre-intervention period, and 1 for post intervention. This
121 variable estimated any abrupt changes detected in the mean concentration (μgm^3) of the air
122 pollutants. Other variables included were: the outcome of interest (monthly mean concentration
123 of air pollutants ($\mu\text{g}/\text{m}^3$)), represented by Y_t ; the base line period (β_0) which provided the mean of
124 the outcome at the beginning of the study period (time=0); time, a continuous variable from the
125 beginning of the study period (2013) to the end (2017); (β_1), which captured the mean monthly
126 trend of air pollutants(μgm^3) before the policy, allowing us to account for long term trends in the
127 outcome of interest over time and an interaction term (β_3) between the policy and time which
128 detected any gradual changes after the policy, in comparison to before. Random variability not
129 explained by the model was represented by ϵ_t (error term at time t)

$$Y_t = \beta_0 + \beta_1 \times \text{time}_t + \beta_2 \times \text{intervention}_t + \beta_3 \times \text{time after intervention}_t + \varepsilon_t \quad (1)$$

The Durbin-Watson statistic test was used to test for serial autocorrelation between the error terms in the model (Wagner et al. 2002), with the Cochrane-Orcutt estimator used to correct for autocorrelation. The validity of the ITS model was further assessed by running three additional models, using three false shipping policy periods: 6 months, 12 months and 24 months before the intervention (Stallings-Smith et al. 2013).

The second method, CUSUM charts, were used to verify that a change had occurred at the time of the policy implementation during the period January 1st, 2010 – June 30th, 2017. The method was developed initially for industrial purposes and recently the technique has been adapted and used to detect changes in air pollution concentrations (Barratt et al. 2007; Carslaw, D.C, Ropkins, K., Bell 2006; Chelani 2011a). In order to apply the technique to detect any changes in the air pollutants for our study, the following procedure was applied: The pre-implementation period (base period), January 1st, 2010 to June 30th, 2015 was used for the calculation of the reference mean and standard deviation. The CUSUM method proposed by Lucas and Crosier (1982) (Barratt et al. 2007) was then applied to individual observations in our time series to calculate the deviations of observations away from the reference mean. Any increase or decrease in mean concentration of the pollutants away from the reference mean were detected by the upper (UDB) and lower (LDB) boundaries on the chart, where the reference value (K) was set to 0.5, as this is the appropriate choice for K in detecting when a 1-sigma shift in the process mean, and it has been shown that sensitivity to detecting shifts away from the mean increases when a smaller reference value is used (Barratt et al. 2007). Confidence limit ($\pm h\sigma_x$) was also set for the CUSUM charts to indicate when a 1-sigma shift in the process mean has occurred (Barratt et al. 2007). The value of the parameter h is generally set to 4 or 5. We selected h=4 as it has been

widely used in previous studies pertaining to air pollution concentration change (Barratt et al. 2007) (Chelani 2011b). Detailed description of the CUSUM method charts has been published elsewhere (Barratt et al. 2007) (Chelani 2011b) (Jones et al. 2012) (Carslaw, D.C, Ropkins, K., Bell 2006). Air pollution concentrations were deseasonalized using the seasonal decomposition time series by Loess (STL) function prior to analysis. All analyses were carried out using the software package R 3.4.2. The R package “qcc” was used for the CUSUM analysis (R Core Team 2016).

2.4. Health impact assessment

The health impact of the implementation of the shipping emission policy was based on a previously established relation between changes in air quality ($10\mu\text{g}\text{m}^{-3}$) and the mortality rate from all natural causes, cardiovascular diseases and respiratory disease in Hong Kong (Lai et al. 2013). Mortality rates for Hong Kong for 2010-2016 were taken from vital statistics(CHP). Excess deaths due to air pollution from SO_2 , PM_{10} NO_2 and O_3 were compared for the periods 2010-2014 versus 2015-2016 as well as 2010 versus 2016. If the excess risk was negative 0 deaths were assumed (Lai et al. 2013), as occurred for O_3 . The model is described in detail in Appendix B.

3. Results

3.1. Air quality changes after implementation of the policy

Table. 1 shows the average ambient concentrations for SO_2 , PM_{10} NO_2 and O_3 at the 10 selected monitoring stations, for two and a half years before and after the policy implementation. The port monitoring station (KC) had the highest levels of SO_2 before the intervention and the largest

reduction in SO₂ concentration after the policy was implemented. Changes in the other pollutants and at other monitoring stations were less marked.

Table 2 shows that the fall in SO₂ at the port monitoring station (KC) when the policy was implemented was abrupt but not gradual. No changes in PM₁₀, NO₂ or O₃ were evident at KC. Table 2 shows no such abrupt (or gradual) change in SO₂ at the non-shipping control stations (TP, YL and TPM). However abrupt declines also occurred at two other monitoring stations close to the port, with a gradual decline at one other station (Appendix C)), although the CUSUM charts indicated the decline might have started before the policy change (Appendix D). No decline in SO₂, PM₁₀, NO₂ or O₃ were evident for Hong Kong although the possibility of a small abrupt change in SO₂ cannot be excluded. After controlling for seasonality, autocorrelation was still present in the error terms based on the Durbin-Watson (DW) statistic test which gave a value of <2. Autocorrelation was then corrected by using the Cochrane-Orcutt estimator. The results were similar. In the sensitivity analysis, the 3 false policy periods we evaluated before the policy showed no significant changes in mean monthly SO₂ concentration compared to the actual policy periods (see supplementary Table 4).

Figure 3, using CUSUM analysis, confirmed a reduction in ambient SO₂ concentration at the shipping port (KC) at the time the shipping emission policy was implemented. In contrast, at one of the control station (TP) SO₂ appears to have been declining before the policy was implemented. For Hong Kong as a whole the CUSUM analysis does not confirm a reduction in ambient SO₂ concentration when the shipping emission policy was implemented (Figure 4).

Similarly, for the other stations in Hong Kong CUSUM showed SO₂ concentration to be declining before implementation of the policy (Appendix D).

3.2. Health impact associated with the policy

Based on the changes in these air pollutants for all of Hong Kong during the period 2010-2014 compared to 2015-2016 the total number of avoidable deaths per year for all natural causes (deaths not caused by external forces), cardiovascular disease and respiratory disease were 379, 72 and 30 respectively for all ambient pollutants combined. The corresponding avoidable deaths per year for all natural causes, cardiovascular disease and respiratory disease attributable to ambient SO₂ were 118, 18 and 10. In comparison to the above time period evaluated, avoidable deaths were higher when comparing 2010 with 2016. Combining the ambient concentrations of all 4 pollutants, avoidable deaths per year for natural causes, cardiovascular disease and respiratory disease were 437, 92 and 36 respectively, with cardiovascular disease having more avoidable deaths than respiratory diseases. The corresponding avoidable deaths associated with SO₂ during this period were also slightly higher for natural cause of death (123), cardiovascular diseases (20) and respiratory diseases (11) (Table 3.).

4. Discussion

Consistent with the implementation of policies to reduce the sulfur content of shipping fuel in Europe and the United States (Schembari, C; Cavalli, F; Cuccia, E; Hjorth 2012)(Velders et al. 2011), this first study of the implementation of a shipping emission policy on the sulfur content of fuel in Hong Kong shows it was very effective in reducing SO₂ levels near the major shipping port in Hong Kong, which previously had the highest levels of SO₂ in Hong Kong. However, the

effects for Hong Kong overall and for Hong Kong excluding the major shipping port were less marked but not at all negligible (e.g., 20% reduction for PM₁₀).

Our study had sufficient observations over time to use ITS to draw conclusions about the magnitude of the impact of the intervention on pollution concentrations independent of any secular trend (Penfold and Zhang 2013)(Pharmd et al. 2002)(Lopez Bernal et al. 2016), which is more reliable than a before and after comparison (Penfold and Zhang 2013). We also used CUSUM to confirm that the timing of the change matched the timing of the intervention (Barratt et al. 2007), because the concentration change was large in comparison to seasonal fluctuations and the long term trend (Barratt et al. 2007).

From a public health perspective, this study shows the benefits of reducing the sulfur content of shipping fuel, although this need to be contextualized within the overall environmental and health costs of any other consequences of this policy. More broadly, this study also raises specific issues about air pollution in Hong Kong. The declining trend of SO₂ concentration at the control station (TP) and for all of Hong Kong, is probably a reflection of tighter power plant regulations (Ng et al. 2013) and introduction of the Second West-East Natural Gas Pipeline in 2013— supplying cleaner fuel to generate electricity (Environmental Protection Department 2016). However, SO₂ concentration at non port areas in Hong Kong, such as Tsuen Wan (TW), Sham Shui Po (SS), Central Western (CW), Tung Chung (TC), TPM and YL were also high, possibly because of close proximity to the major shipping lanes in Hong Kong and for the latter two, close location to shipping ports based in mainland China (TPM and YL)(Ng et al. 2013) — which does not have strict marine vessels emission regulations. Although the measures such as tighter power plant regulations, mentioned, might have resulted in synergistic declines of SO₂ in Hong Kong, these interventions were implemented over a long time period for

241 gradual SO₂ reduction in Hong Kong. The shipping emission policy, however, was implemented
242 punctually on July 1st 2015, which is a more likely the explanation for the abrupt decline in
243 SO₂ concentration.

244 SO₂, unlike all other pollutants, has a peak season both in the summer (July and August) and
245 winter months (October and January) (Lau et al. 2005). In the winter months, the pollutant
246 ranges for moderate to high at all stations due to the northwesterly winds which bring in
247 pollution from the Pearl River Delta (Lau et al. 2005). However, during the summer month peak
248 period, the wind direction is from weak south westerlies and stations such as Tsuen Wan (TW),
249 Sham Shui Po (SS), Central Western (CW) and KC in particular, has the highest levels of SO₂
250 with the south westerlies, while other stations the levels are low (Lau et al. 2005). This highlights
251 the significance of SO₂ from local shipping emission sources —such as the large container port
252 located to the south-south west of KC (Lau et al. 2005).

253 The policy undoubtedly reduced air pollution related deaths in Hong Kong, as seen in other
254 similar intervention studies focusing on sulfur and coal bans, which revealed a substantial
255 reduction in SO₂ concentration accompanied by a drop in cardiovascular and respiratory diseases
256 (Clancy et al. 2002)(Hedley et al. 2002). An intervention study carried out by Hedley and
257 company, showed that the 1990 intervention in Hong Kong to reduce sulfur in fuels had resulted
258 in an reduction in seasonal deaths for cardiovascular and respiratory diseases, 12 months after
259 the implementation of the policy (Hedley et al. 2002). Albeit, some research have shown
260 respirable particles to be the main component in air pollution that causes death, some time series
261 studies have reported strong associations between SO₂ and daily cardiorespiratory deaths even
262 after adjusting for other gaseous pollutants and suspended particles (Kan, Haidong; Wong, Chit-
263 Ming; Vichit-Vadakan 2010; Katsouyanni et al. 1997a; Venneris et al. 2003; Wong et al. 2001).

264

265 Our study has some limitations. First, the health impact assessment should be interpreted with
266 caution because it was based on ambient air pollution concentrations and not on concentrations
267 solely due to shipping emissions. For the health evaluation, we also adopted a method which is a
268 simple, robust and conservative (Hedley et al. 2008). Nevertheless, the total estimate of
269 avoidable deaths is close to what was predicted for a 0.5% reduction in sulfur content of marine
270 vessel fuel in Hong Kong (Kilburn et al. 2012). Moreover, many studies have also indicated an
271 association of SO₂ with hospital admissions and deaths (Chen et al. 2012; Derriennic et al. 1989;
272 Geravandi et al. 2016; Kan et al. 2010; Kermani et al. 2016; Wong et al. 2002).

273 Second, we estimated health effects across Hong Kong from levels at the monitoring stations
274 rather than on individual exposures. However, Hong Kong is very compact and most people live
275 within close proximity to the monitoring stations (Huang et al. 2017) . Third, due to the lack of
276 data we were unable to evaluate PM₁₀ components that are specific to shipping emissions.

277 However, in Hong Kong, PM₁₀ comes from 4 major sources: traffic, local power plants, regional
278 air pollution from China and ships. Out of these 4 sources, shipping emission only accounts for
279 a small fraction (1/3). Lastly, the CUSUM technique was used in the most basic form, which
280 means any subtle changes in pollution levels were not detected because underlying temporal
281 variation was not adjusted for in the mean and standard deviation used for the CUSUM plots.

282 **5. Conclusion**

283 Implementation of a policy to reduce the sulfur content of shipping fuel substantially reduced
284 ambient SO₂ with likely corresponding effects on deaths. However, future research on the effect
285 source specific shipping emission pollutants have on mortality is needed, to estimate more

accurate mortality rates associated with shipping emissions. For coastal cities like Hong Kong to gain full benefits, the policy needs to be implemented regionally.

6. Acknowledgements

We thank the Center for Health Protection (CHP) report for supplying the health data and the Environmental Protection Department of the Hong Kong Special Administrative Region for providing the air monitoring data for this study.

References

- Barratt B, Atkinson R, Ross Anderson H, Beevers S, Kelly F, Mudway I, et al. 2007. Investigation into the use of the CUSUM technique in identifying changes in mean air pollution levels following introduction of a traffic management scheme. *Atmos. Environ.* 41:1784–1791; doi:10.1016/j.atmosenv.2006.09.052.
- Chen R, Huang W, Wong C-M, Wang Z, Thach Q, Chen B, et al. 2012. Short-term exposure to sulfur dioxide and daily mortality in 17 Chinese cities: The China air pollution and health effects study (CAPES). *Environ. Res.* 118:101–106; doi:10.1016/j.envres.2012.07.003.
- CHP. Centre for Health Protection - Age-standardised Death Rates by Leading Causes of Death, 2001 - 2016. Available: <http://www.chp.gov.hk/en/data/4/10/27/339.html> [accessed 2 November 2017].

307 Clancy L, Goodman P, Sinclair H, Dockery DW. 2002. Effect of air-pollution control on death
308 rates in Dublin, Ireland: an intervention study. *Lancet* 360:1210–1214;
309 doi:[http://dx.doi.org/10.1016/S0140-6736\(02\)11281-5](http://dx.doi.org/10.1016/S0140-6736(02)11281-5).

310 Corbett, J J, Winerbrake, J.J., Green, E.H. K. 2007. Mortality from Ship Emissions: A Global
311 Assessment. *Environmen. Sci. technol.* 41:8512–8518; doi:10.1021/es071686z.

312 Dennis J, Ramsay T, Turgeon AF, Zarychanski R. 2013. Helmet legislation and admissions to
313 hospital for cycling related head injuries in Canadian provinces and territories: interrupted
314 time series analysis. *BMJ* 346:f2674; doi:10.1136/BMJ.F2674.

315 Derriennic F, Richardson S, Mollie A, Lellouch J. 1989. Short-term effects of sulphur dioxide
316 pollution on mortality in two French cities. *Int. J. Epidemiol.* 18: 186–97.

317 Dockery DW, Rich DQ, Goodman PG, Clancy L, Ohman-Strickland P, George P, et al. 2013.
318 Effect of air pollution control on mortality and hospital admissions in Ireland. *Res. Rep.*
319 *Health. Eff. Inst.* 3–109.

320 Endresen Ø, Sørgård E, Sundet JK, Dalsøren SB, Isaksen ISA, Berglen TF, et al. 2003. Emission
321 from international sea transportation and environmental impact. *J. Geophys. Res.* 108:4560;
322 doi:10.1029/2002JD002898.

323 Environmental Protection Department. 2016. AQO Review Working Group Energy and Power
324 Generation Sub-group Emission Control Measures on Power Generation and Energy
325 Efficiency E&PG Paper 2/2016 For information.

326 Environmental Protection Department. 2015. Ocean Going Vessels Fuel Switch At Berth.

327 Eyring V, Köhler HW, van Aardenne J, Lauer A. 2005. Emissions from international shipping: 1.

328 The last 50 years. *J. Geophys. Res.* 110:D17305; doi:10.1029/2004JD005619.

329 Geravandi S, Goudarzi G, Soltani F, Salmanzadeh S, Ghomeishi A, Zalaghi E, et al. 2016. The
330 cardiovascular and respiratory deaths attributed to sulfur dioxide in Kermanshah. *J.*
331 Kermanshah Univ. Med. Sci. 19: 319–326.

332 Hedley AJ, McGhee SM, Barron B, Chau P, Chau J, Thach TQ, et al. 2008. Air pollution: costs
333 and paths to a solution in Hong Kong--understanding the connections among visibility, air
334 pollution, and health costs in pursuit of accountability, environmental justice, and health
335 protection. *J Toxicol Env. Heal. A* 71:544–554; doi:10.1080/15287390801997476.

336 Hedley AJ, Wong CM, Thach TQ, Ma S, Lam TH, Anderson HR. 2002. Cardiorespiratory and
337 all-cause mortality after restrictions on sulphur content of fuel in Hong Kong: An
338 intervention study. *Lancet*; doi:10.1016/S0140-6736(02)11612-6.

339 Huang J V, Leung GM, Schooling CM. 2017. The Association of Air Pollution With Pubertal
340 Development: Evidence From Hong Kong's "Children of 1997" Birth Cohort.
341 *Am. J. Epidemiol.* 185; doi:10.1093/aje/kww200.

342 International Maritime Organization. 2015. HIGH-LEVEL ACTION PLAN OF THE
343 ORGANIZATION AND PRIORITIES FOR THE 2016-2017 BIENNIUM. Available:
344 [http://www.imo.org/en/About/strategy/Documents/A 29-Res.1098 - HLAP 2016-2017.pdf](http://www.imo.org/en/About/strategy/Documents/A%20Res.1098%20-%20HLAP%202016-2017.pdf)
345 [accessed 8 December 2017].

346 Kan, Haidong; Wong, Chit-Ming; Vichit-Vadakan N. 2010. Short-term association between
347 sulfur dioxide and daily mortality: The Public Health and Air Pollution in Asia (PAPA)
348 study. *Environ. Res.* 110:258–264; doi:10.1016/J.ENVRES.2010.01.006.

349 Kan H, Wong C-M, Vichit-Vadakan N, Qian Z, Teams the P project. 2010. Short-term
 350 association between sulfur dioxide and daily mortality The Public Health and Air Pollution
 351 in Asia (PAPA) study. *Environ. Res.* 110:258–264; doi:10.1016/j.envres.2010.01.006.

352 Katsouyanni K, Touloumi G, Spix C, Schwartz J, Balducci F, Medina S, et al. 1997a. Short-term
 353 effects of ambient sulphur dioxide and particulate matter on mortality in 12 European cities:
 354 results from time series data from the APHEA project. *Air Pollution and Health: a European*
 355 *Approach.* *BMJ* 314:1658–63; doi:10.1136/BMJ.314.7095.1658.

356 Katsouyanni K, Touloumi G, Spix C, Schwartz J, Balducci F, Medina S, et al. 1997b. Short-term
 357 effects of ambient sulphur dioxide and particulate matter on mortality in 12 European cities:
 358 results from time series data from the APHEA project. *Air Pollution and Health: a European*
 359 *Approach.* *Bmj* 314: 1658–1663.

360 Kermani M, Fallah Jokandan S, Aghaei M, Bahrami Asl F, Karimzadeh S, Dowlati M. 2016.
 361 Estimation of the Number of Excess Hospitalizations Attributed to Sulfur Dioxide in Six
 362 Major Cities of Iran. *Heal. Scope* 5; doi:10.17795/jhealthscope-38736.

363 Kilburn M, Ng SKW, Chubin L, Chan J, Yip A, Lau AKH, et al. 2012. A Price Worth Paying:
 364 The Case for Controlling Marine Emissions in the Pearl River Delta Civic Exchange About
 365 the Organisations.

366 Lai HK, Tsang H, Chau J, Lee CH, McGhee SM, Hedley AJ, et al. 2013. Health impact
 367 assessment of marine emissions in Pearl River Delta region. *Mar. Pollut. Bull.* 66:158–163;
 368 doi:10.1016/j.marpolbul.2012.09.029.

369 Lau K-H, Wu WM, Fung JCH, Henry RC, Barron B. 2005. Significant Marine Source for SO₂
 370 levels in Hong Kong.

371 Ng S. 2013. Cruise Ship Emissions and Control in Hong Kong About Civic Exchange.

372 Ng SKW, Loh C, Lin C, Booth V, Chan JWM, Yip ACK, et al. 2013. Policy change driven by an
 373 AIS-assisted marine emission inventory in Hong Kong and the Pearl River Delta. *Atmos.*
 374 *Environ.* 76:102–112; doi:http://dx.doi.org/10.1016/j.atmosenv.2012.07.070.

375 Nistal-Nuño B. 2017. Segmented regression analysis of interrupted time series data to assess
 376 outcomes of a South American road traffic alcohol policy change. *Public Health* 150:51–59;
 377 doi:10.1016/J.PUHE.2017.04.025.

378 North-West Ports Clean Air Strategy. Port of Seattle, Port of Tacoma, Vancouver Port Authority.
 379 Available:
 380 [https://www.nwseaportalliance.com/sites/default/files/final_2015_implementation_report_v](https://www.nwseaportalliance.com/sites/default/files/final_2015_implementation_report_version_3-14-2017.pdf)
 381 [ersion_3-14-2017.pdf](https://www.nwseaportalliance.com/sites/default/files/final_2015_implementation_report_version_3-14-2017.pdf) [accessed 8 December 2017].

382 Penfold RB, Zhang F. 2013. Use of Interrupted Time Series Analysis in Evaluating Health Care
 383 Quality Improvements. *ACAP* 13:S38–S44; doi:10.1016/j.acap.2013.08.002.

384 Pharmd AKW, Scd SBS, Ms FZ, Scd DR, Wagner a. K, Soumerai SB, et al. 2002. Segmented
 385 regression analysis of interrupted time series studies in medication use research. *J. Clin.*
 386 *Pharm. Ther.* 27:299–309; doi:10.1046/j.1365-2710.2002.00430.x.

387 San Pedro Bay Ports Clean Air Action Plan. The Port of Los Angeles | Environment. Available:
 388 <https://www.portoflosangeles.org/environment/caap.asp> [accessed 8 December 2017].

389 Schembari, C; Cavalli, F; Cuccia, E; Hjorth J. 2012. Impact of a European directive on ship
 390 emissions on air quality in Mediterranean harbours. *Atmos. Environ.* 61:661–669;
 391 doi:10.1016/J.ATMOSENV.2012.06.047.

392 Stallings-Smith S, Zeka A, Goodman P, Kabir Z, Clancy L. 2013. Reductions in Cardiovascular,
 393 Cerebrovascular, and Respiratory Mortality following the National Irish Smoking Ban:
 394 Interrupted Time-Series Analysis. L. Manzoli, ed PLoS One 8:e62063;
 395 doi:10.1371/journal.pone.0062063.

396 Stieb DM, Judek S, Burnett RT. 2002. Meta-Analysis of Time-Series Studies of Air Pollution
 397 and Mortality: Effects of Gases and Particles and the Influence of Cause of Death, Age, and
 398 Season. *J. Air Waste Manage. Assoc.* 52:470–484; doi:10.1080/10473289.2002.10470794.

399 Stieb DM, Judek S, Burnett RT. 2003. Meta-Analysis of Time-Series Studies of Air Pollution
 400 and Mortality: Update in Relation to the Use of Generalized Additive Models. *J. Air Waste*
 401 *Manage. Assoc.* 53:258–261; doi:10.1080/10473289.2003.10466149.

402 Velders GJM, Snijder A, Hoogerbrugge R. 2011. Recent decreases in observed atmospheric
 403 concentrations of SO₂ in the Netherlands in line with emission reductions. *Atmos. Environ.*
 404 45:5647–5651; doi:10.1016/j.atmosenv.2011.07.009.

405 Venners SA, Wang B, Peng Z, Xu Y, Wang L, Xu X. 2003. Particulate Matter, Sulfur Dioxide,
 406 and Daily Mortality in Chongqing, China. *Environ. Health Perspect.* 111:562–567;
 407 doi:10.1289/ehp.5664.

408 Viana, M.; Hammingh, P.; Colette, A.; Querol X. 2014. Impact of maritime transport emissions
 409 on coastal air quality in Europe. *Atmos. Environ.* 90:96–105;
 410 doi:10.1016/J.ATMOSENV.2014.03.046.

411 Wagner AK, Soumerai SB, Zhang F, Ross-Degnan D. 2002. Segmented regression analysis of
 412 interrupted time series studies in medication use research. *J. Clin. Pharm. Ther.* 27:299–
 413 309; doi:10.1046/j.1365-2710.2002.00430.x.

414 Winebrake JJ, Corbett JJ, Green EH, Lauer A, Eyring V. 2009. Mitigating the health impacts of
 415 pollution from oceangoing shipping: an assessment of low-sulfur fuel mandates. *Env. Sci*
 416 *Technol* 43: 4776–4782.

417 Wong C-M, Atkinson RW, Ross Anderson H, Johnson Hedley A, Ma S, Yuen-Kwan Chau P, et
 418 al. 2002. A Tale of Two Cities: Effects of Air Pollution on Hospital Admissions in Hong
 419 Kong and London Compared. *Environ. Heal. Perspect.* • 110.

420 Wong C-M, Ma S, Hedley AJ, Lam T-H. 2001. Effect of Air Pollution on Daily Mortality in
 421 Hong Kong. *Environ. Health Perspect.* 109.

422 Wong CM, Lam TH, Peters J, Hedley AJ, Ong SG, Tam AY, et al. 1998. Comparison between
 423 two districts of the effects of an air pollution intervention on bronchial responsiveness in
 424 primary school children in Hong Kong. *J. Epidemiol. Community Health* 52:571–8;
 425 doi:10.1136/JECH.52.9.571.

426 Yau PS, Lee SC, Cheng Y, Huang Y, Lai SC, Xu XH. 2013. Contribution of ship emissions to
 427 the fine particulate in the community near an international port in Hong Kong. *Atmos. Res.*
 428 124:61–72; doi:http://dx.doi.org/10.1016/j.atmosres.2012.12.009.

429 Yau PS, Lee SC, Corbett JJ, Wang C, Cheng Y, Ho KF. 2012. Estimation of exhaust emission
 430 from ocean-going vessels in Hong Kong. *Sci. Total Environ.* 431:299–306;
 431 doi:http://dx.doi.org/10.1016/j.scitotenv.2012.03.092.

432

433

Figures

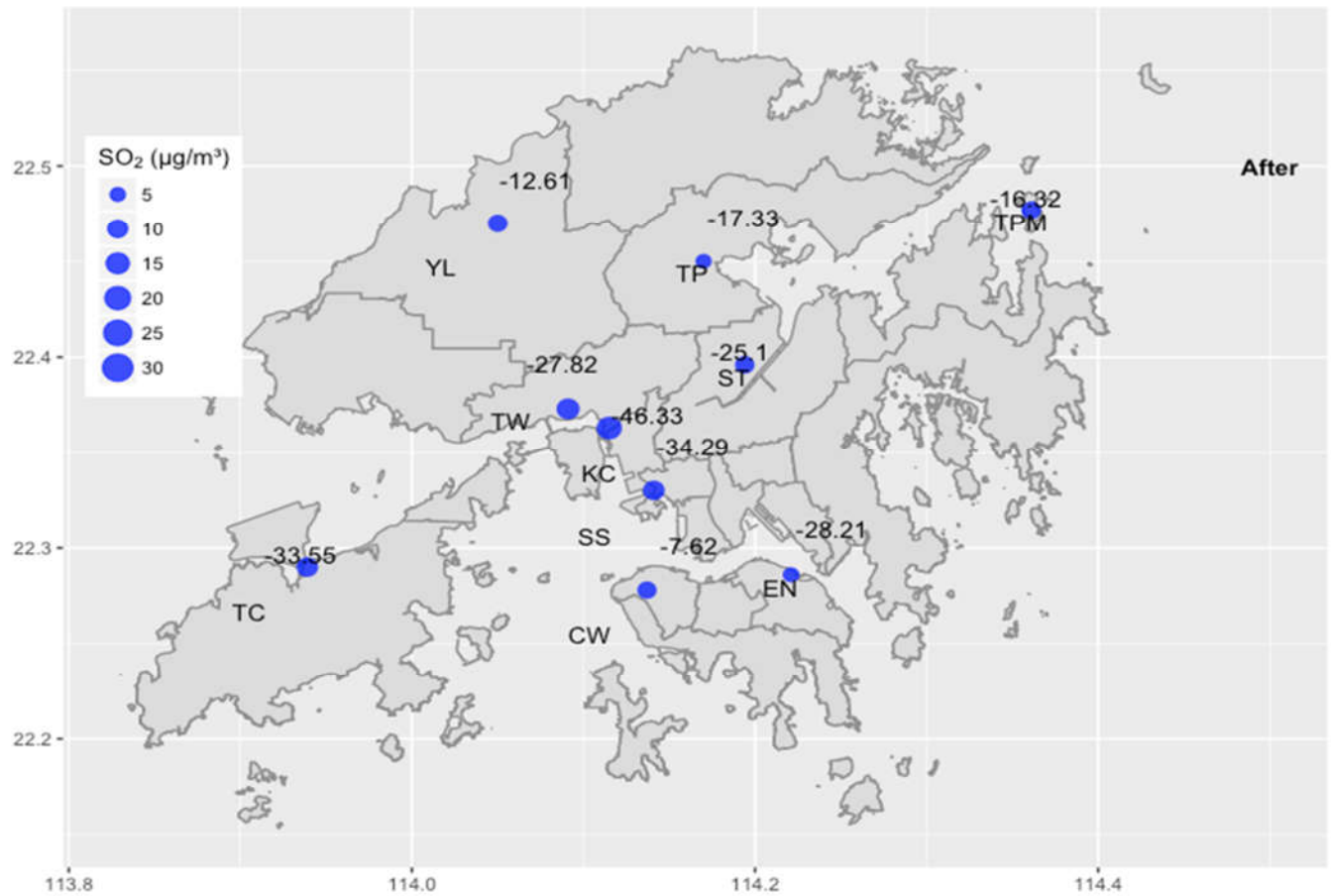


Fig.1. Spatial map of Hong Kong showing SO₂ declining percent concentration change, at the 10 monitoring stations, 2 years after the shipping emission policy.

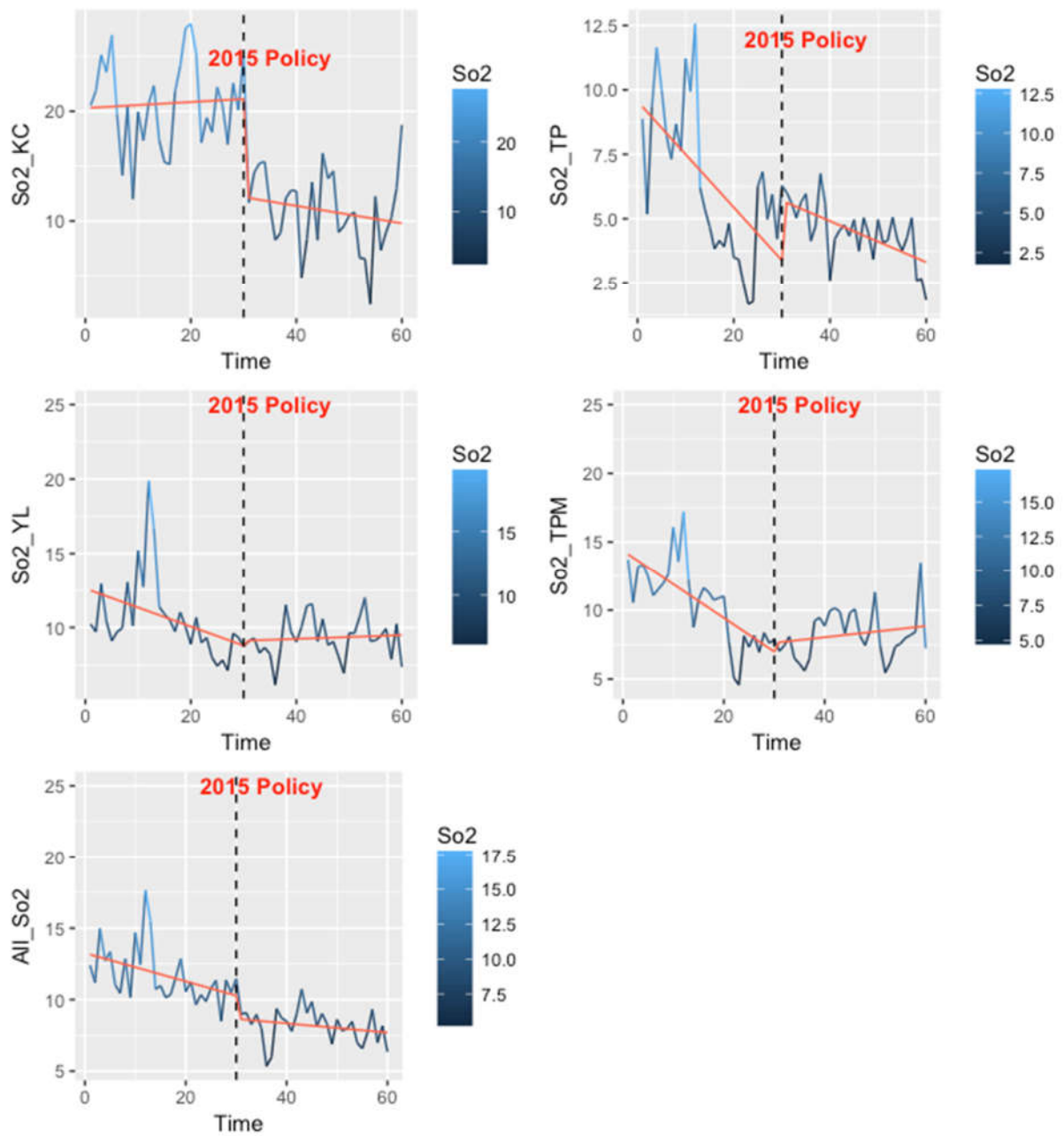


Fig. 2. Time series plot of monthly mean concentrations of SO₂ at KC, TP, YL, TPM, and all of Hong Kong.

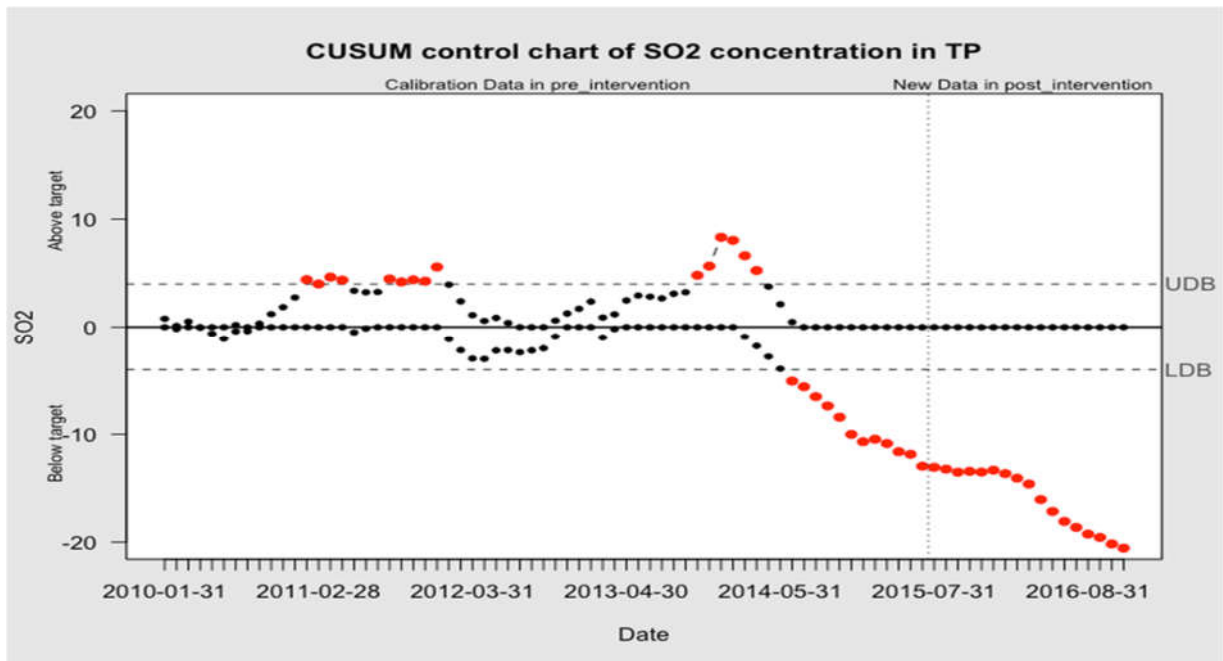
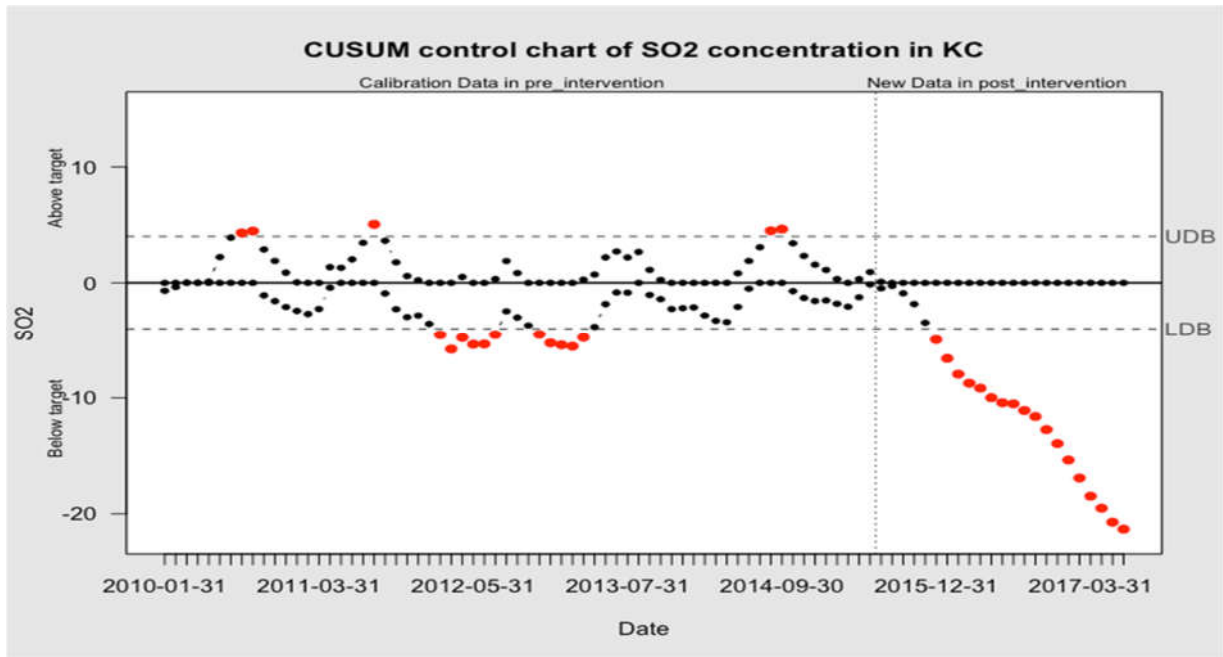


Fig.3. CUSUM chart comparing KC and TP SO₂ concentrations before and after the shipping emission policy.

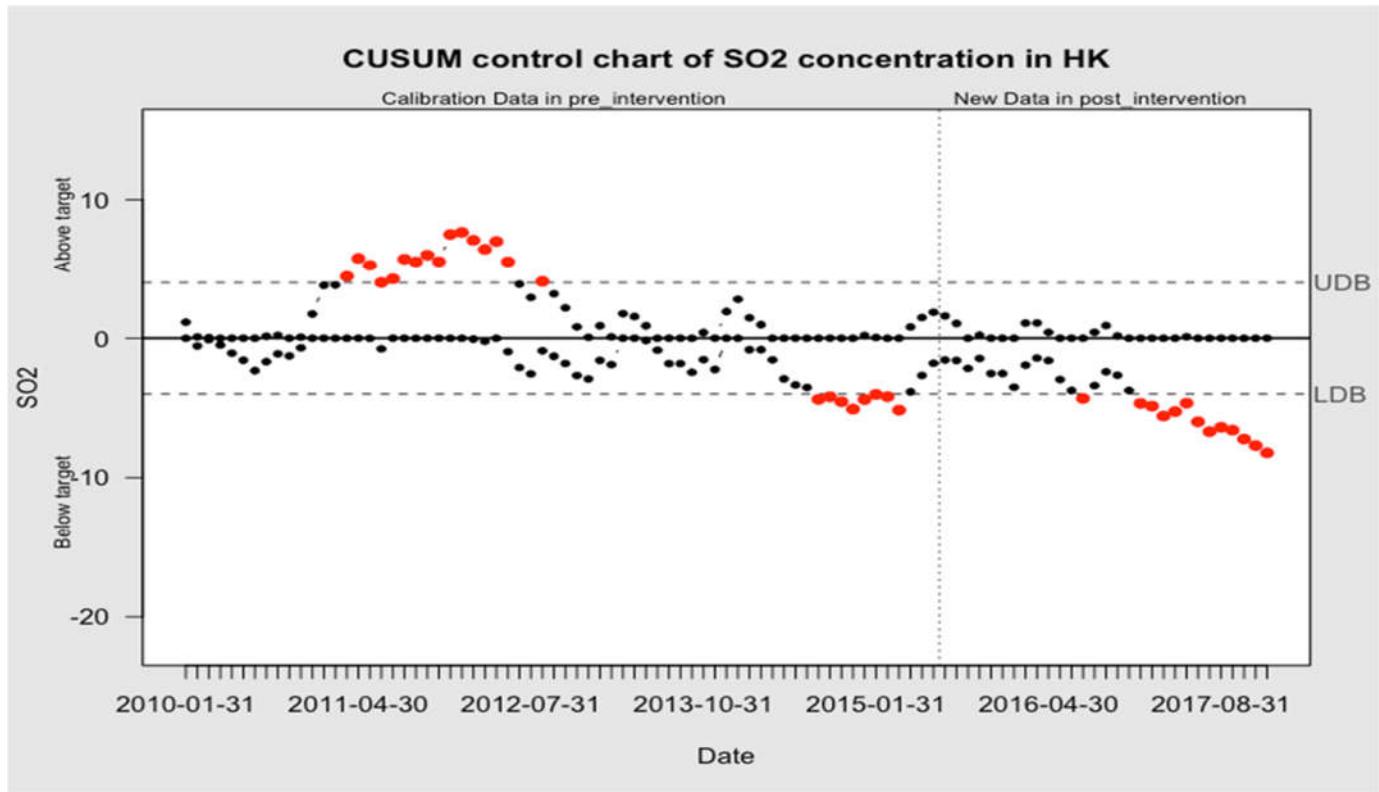


Fig.4. CUSUM chart comparing SO₂ concentrations before and after the shipping emission for all of Hong Kong.

Tables

Table 1. Mean concentrations of SO₂, PM₁₀, NO₂ and O₃ at the 10 general monitoring stations 2.5 years before and after implementation of the shipping emission policy (2013-2017).

Monitoring stations		SO ₂ (µgm ³)			PM ₁₀ (µg m ³)			NO ₂ (µgm ³)			O ₃ (µgm ³)		
Location	Name	Pre	Post	%	Pre	Post	%	Pre	Post	%	Pre	Post	%
Port	KC	21.0	11.0	-50.0	42.0	34.0	-20.0	68.0	58.0	-14.0	33.0	37.0	10.0
Control	TP	5.9	4.6	-21.0	41.0	32.0	-23.0	46.0	36.0	-22.0	49.0	49.0	-0.57
	YL	11.0	9.3	-12.0	52.0	39.0	-25.0	52.0	43.0	-17.0	37.0	40.0	8.8
	TPM	10.0	8.2	-21.0	45.0	33.0	-28.0	11.0	10.0	-8.9	74.0	72.0	-2.4
Other	TW	16.0	11.0	-29.0	43.0	32.0	-25.0	60.0	46.0	-24.0	35.0	38.0	8.2
	CW	9.9	8.9	-11.0	46.0	33.0	-27.0	50.0	42.0	-17.0	41.0	48.0	17.0
	EN	6.6	4.6	-31.0	40.0	31.0	-21.0	54.0	44.0	-19.0	40.0	54.0	34.0
	SS	14.0	8.7	-37.0	44.0	34.0	-24.0	68.0	56.0	-17.0	33.0	37.0	12.0
	ST	10.0	7.1	-29.0	40.0	30.0	-25.0	44.0	38.0	-15.0	49.0	48.0	-1.9
	TC	13.0	8.6	-33.0	41.0	33.0	-18.0	46.0	37.0	20.0	45.0	45.0	0.13

Table 2. Estimates of the abrupt and gradual changes in SO₂, PM₁₀, NO₂ and O₃ at the port (KC) monitoring station after implementation of the shipping emission policy.

Station	Air Pollutant	change	Beta	se	t-values	p-values
KC	SO ₂ (μgm ³)	Abrupt	-10.0	2.7	-3.8	0.0004
		Gradual	-0.081	0.17	-0.46	0.65
	PM ₁₀ (μgm ³)	Abrupt	-7.8	5.4	-1.4	0.16
		Gradual	0.54	0.34	1.6	0.12
	NO ₂ (μgm ³)	Abrupt	-4.8	4.9	-0.98	0.33
		Gradual	0.004	0.31	0.014	0.99
	O ₃ (μgm ³)	Abrupt	-1.5	4.7	-0.33	0.74
		Gradual	-0.31	0.28	-1.1	0.28
All of HK						
	SO ₂ (μgm ³)	Abrupt	-1.6	0.91	-1.8	0.084
		Gradual	0.071	0.055	1.3	0.20
	PM ₁₀ (μgm ³)	Abrupt	3.6	4.6	0.77	0.44
		Gradual	0.14	0.28	0.52	0.61
	NO ₂ (μgm ³)	Abrupt	5.4	2.7	2.0	0.049
		Gradual	-0.24	0.16	-1.5	0.14
	O ₃ (μgm ³)	Abrupt	0.92	3.8	0.25	0.81
		Gradual	0.15	0.23	0.64	0.52
Control						
TP	SO ₂ (μgm ³)	Abrupt	1.6	1.3	1.3	0.21
		Gradual	0.12	0.092	1.3	0.20
YL	SO ₂ (μgm ³)	Abrupt	0.62	1.5	0.43	0.67
		Gradual	0.16	0.099	1.6	0.11
TPM	SO ₂ (μgm ³)	Abrupt	0.37	1.4	0.27	0.79
		Gradual	0.28	0.092	3.1	0.003

Table 3. Estimates of avoided deaths due to the decline of four ambient pollutants ($\mu\text{g}/\text{m}^3$) after the shipping emission policy, comparing two time periods: 1) 2010-2014 vs.2015-2016 & 2) 2010 vs.2016

<i>Excess deaths</i>	<i>Air pollutant</i> (μgm^3)	N_1	ER_p		L_p		Avoidable deaths	
		2010-2014 vs. 2015-2016	2010 vs. 2016	2010 - 2016	2010-2014 vs. 2015-2016	2010 vs. 2016	2010-2014 vs. 2015-2016	2010 vs. 2016
<i>All natural causes (T)</i>	SO ₂	41777	40835	0.71	4.0	4.2	118	123
	NO ₂	41777	40835	1.4	7.4	8.8	431	501
	PM ₁₀	41777	40835	0.31	11.0	14.0	145	181
	O ₃	41777	40835	0.42	-0.95	-1.2	-17	-20
<i>total</i>							379	437
<i>Cardiovascular disease</i>	SO ₂	6298	6636	0.72	4.0	4.2	18	20
	NO ₂	6298	6636	1.6	7.4	8.8	75	94
	PM ₁₀	6298	6636	0.49	11.0	14.0	35	46
	O ₃	6298	6636	0.51	-0.95	-1.2	-3	2
<i>Total</i>							72	92
<i>respiratory disease</i>	SO ₂	1904	2093	1.3	4.0	4.2	10	11
	NO ₂	1904	2093	2.2	7.4	8.8	31	40
	PM ₁₀	1904	2093	0.57	11.0	14.0	12	17
	O ₃	1904	2093	0.48	-0.95	-1.2	-1	-1
<i>Total</i>							30	36

N_1 is the annual number of the event I in the population

ER_p is the pooled excess risk (%) for $10\mu\text{g}/\text{m}^3$ change in pollutant

L_p the change in the level of pollutant P for the reduction from one defined year to the other

P is each of the criteria pollutant.

* $T = \text{SO}_2 + 0.88\text{PM}_{10} + 0.31\text{NO}_2 + \text{O}_3$

Supplementary and Appendix

Supplementary

S Table 1. Estimates of abrupt and gradual changes in SO₂, at the KC monitoring station at three false policy periods (6 months, 12 months and 24 months) before the original policy date.

Air Pollutant	Change	beta	se	t-value	p-value
SO ₂ (µgm ³) 6 months	Abrupt	1.3	3.2	-0.40	0.70
	Gradual	0.20	0.29	-0.81	0.42
SO ₂ (µgm ³) 12 months	Abrupt	3.4	3.3	1.0	0.30
	Gradual	-0.096	0.35	-0.28	0.78
SO ₂ (µgm ³) 24 months	Abrupt	-3.0	4.1	-0.72	0.50
	Gradual	0.73	2.4	0.30	0.77

Appendix

Appendix A

Segmented regression analysis is the statistical analysis used for the ITS design. With this Statistical analysis, time period is separated into pre and post policy (two segments), Resulting in separate intercepts and slopes estimates in both pre and post period.

Statistical test of changes of slopes and intercept are then carried out pre to post policy Period in order to evaluate the policy impact

The model can also be used to identify if any changes are abrupt or gradual. An abrupt change is represented by a change in level (LC) in the segmented regression models and a change in the trend (TC) represents a gradual change after an intervention, as given in the model below for one change point.

$$Y_t = \beta_0 + \beta_1 \times \text{time}_t + \beta_2 \times \text{intervention}_t + \beta_3 \times \text{time after intervention}_t + \varepsilon_t \quad (1)$$

Mean monthly pollution concentration (μgm^3) is represented by Y_t ; time is the continuous variable specifying time in months at the time t from the beginning of the observation period; intervention is an indicator for time t occurring before (intervention = 0) or after (intervention = 1) the policy, which was implemented at 1 time point in the series (30); time after intervention is a continuous variable counting the number of months after the intervention at time t , coded 0 before the policy and time-66 after the policy (Wagner et al. 2002). In the model, β_0 estimates the baseline level of the mean pollution concentration (μgm^3) per month, at time zero; β_1 estimates the base line trend which is the month to month mean pollution concentration change (μgm^3) per unit time before the policy; β_2 estimates any abrupt changes (LC) in the mean monthly pollution concentration (μgm^3) after intervention, that is, from the end of the preceding segment; and β_3 estimates any gradual changes (TC) in pollution concentration after the policy in comparison to before (Wagner et al. 2002).

Appendix B

The impact of avoidable deaths as a result of the policy was calculated for each pollutant using the equation below.

$$I_p = N_I \times ER_p \times L_p \quad (2) \quad (\text{Hedley et al. 2008})$$

N_I is the annual number of the event I in the population all natural causes: ER_p is the excess risk for $10\mu\text{g}/\text{m}^3$ of pollutant (Lai et al. 2013); L_p the change in the level of pollutant P for the reduction from one defined year to the other: and P is for each of the criteria pollutant (Hedley et al. 2008). Since SO_2 is the dominant pollutant for shipping emissions, equation 3 is the most suitable equation used in order to determine the total number of avoidable

deaths that results from the reduction of SO₂. Values of the pollutants to substitute into the equation were derived from equation 2.

$$T = SO_2 + 0.88PM_{10} + 0.31NO_2 + O_3 \quad (3)$$

This equation accounts for correlations as well as partial correlations between the four pollutants in order to acquire the total health impact (T) (Lai et al. 2013)(Hedley et al. 2008). Excess risks (ERp) for each 10-ug/m³ increase in air pollutants were retrieved from a previous published local study(Lai et al. 2013) and N_I (the annual number of the event I in the population) was obtained from the CHP .

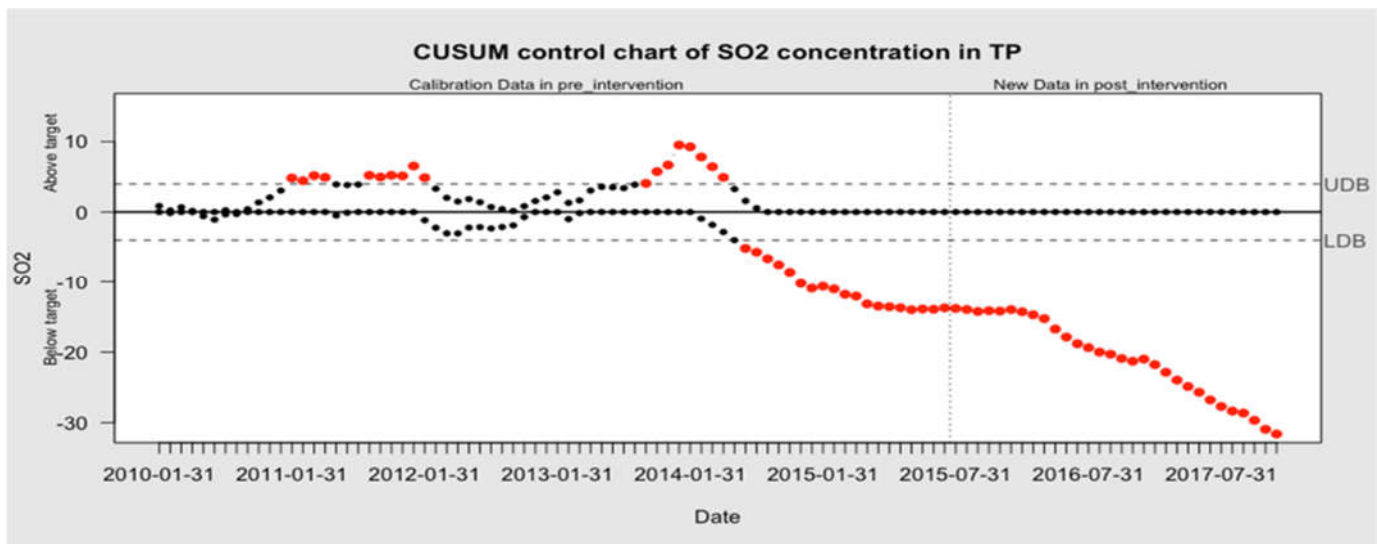
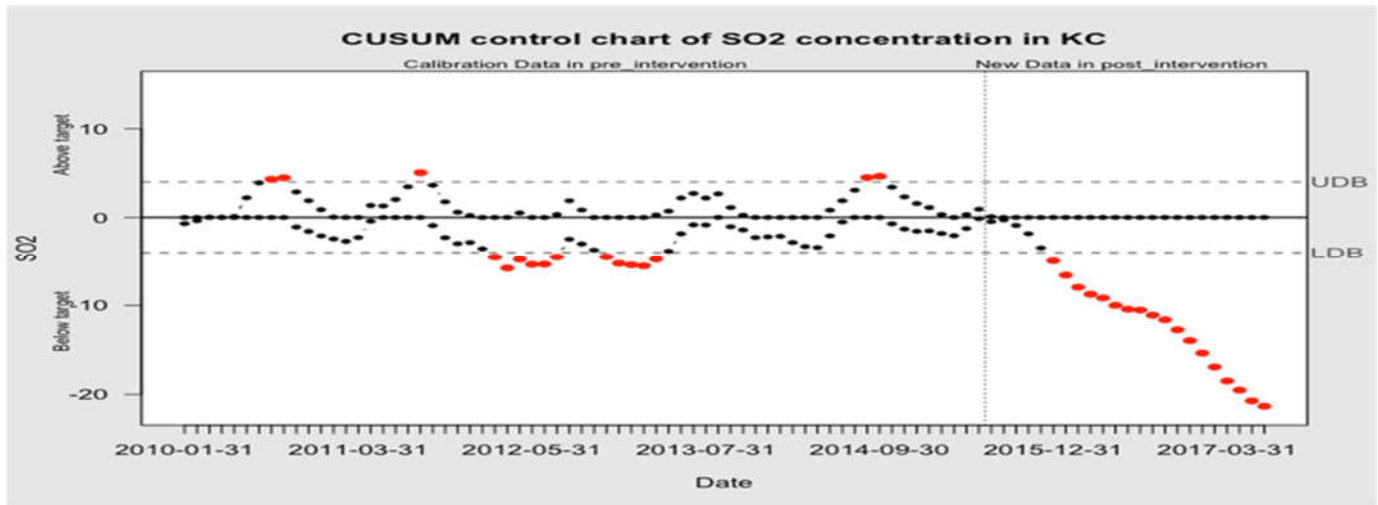
Appendix C

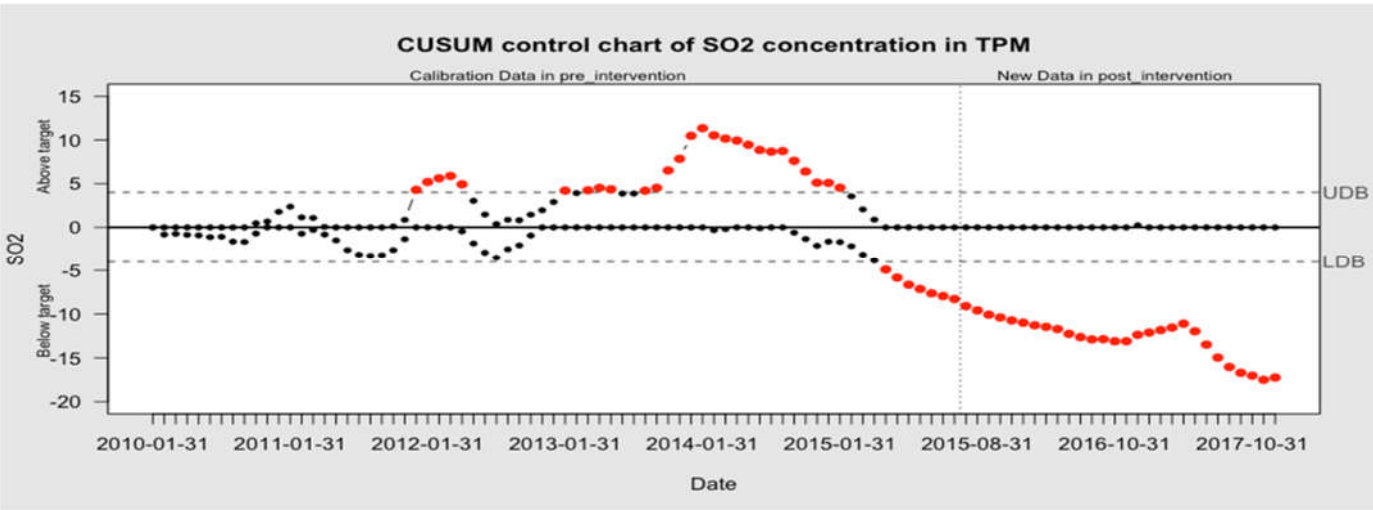
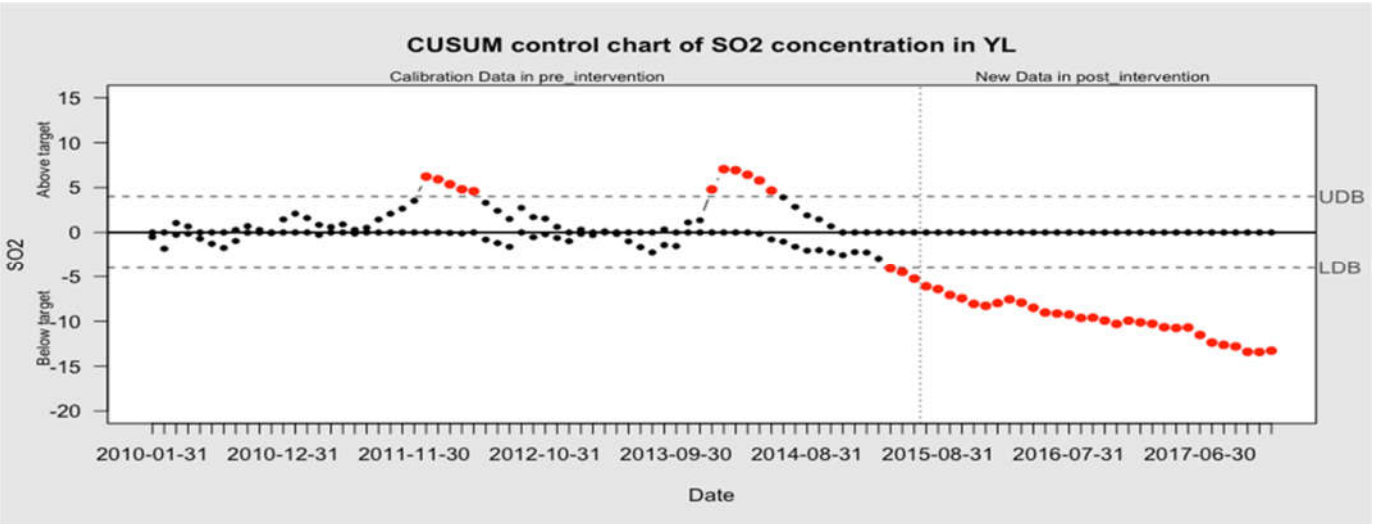
Table 1. Estimates of the abrupt and gradual changes in SO₂ (µgm³), at all 10 monitoring stations in HK after the implementation of the shipping emission policy

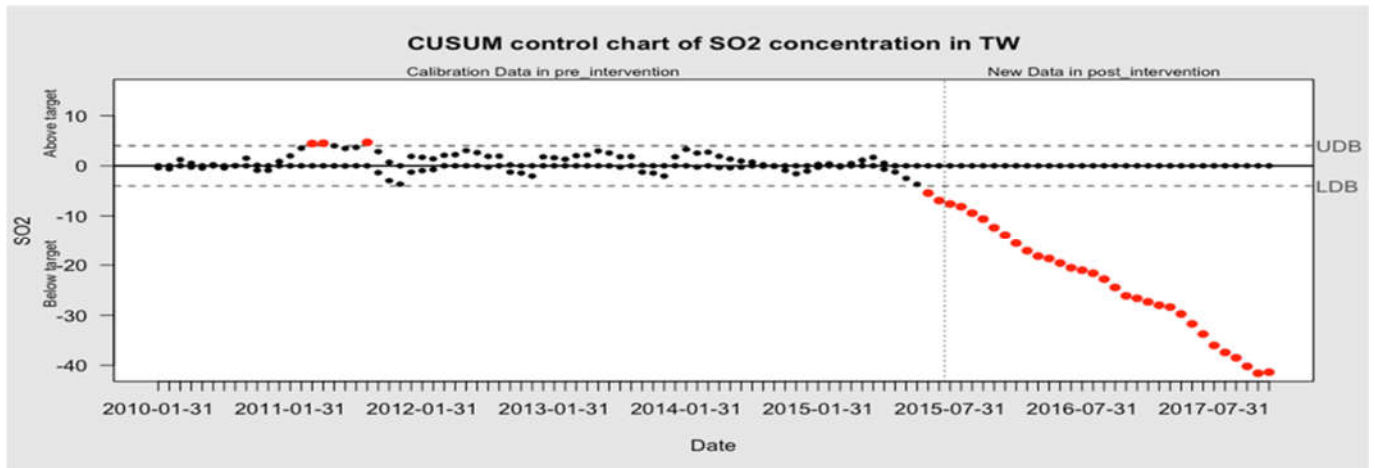
Station	change	Beta	se	t-values	p-values
KC	Abrupt	-10.0	2.7	-3.8	0.0004
	Gradual	-0.081	0.17	-0.46	0.65
TP	Abrupt	1.6	1.3	1.3	0.21
	Gradual	0.12	0.092	1.3	0.20
YL	Abrupt	0.62	1.5	0.43	0.67
	Gradual	0.16	0.099	1.6	0.11
TPM	Abrupt	0.37	1.4	0.27	0.79
	Gradual	0.28	0.092	3.1	0.003
TW	Abrupt	-4.0	1.3	-3.14	0.0027
	Gradual	-0.00036	0.084	-0.004	0.99
CW	Abrupt	-0.98	1.5	-0.65	0.52
	Gradual	-0.026	0.091	-0.28	0.78
EN	Abrupt	-0.79	1.1	-0.71	0.48
	Gradual	0.018	0.070	0.26	0.80
SS	Abrupt	-3.45	1.6	-2.2	0.031
	Gradual	0.080	0.10	0.82	0.41
ST	Abrupt	-1.0	0.85	-1.2	0.25
	Gradual	0.17	0.05	3.3	0.0017

TC	Abrupt	-1.3	2.3	-0.57	0.57
	Gradual	0.27	0.22	1.2	0.23

Appendix D



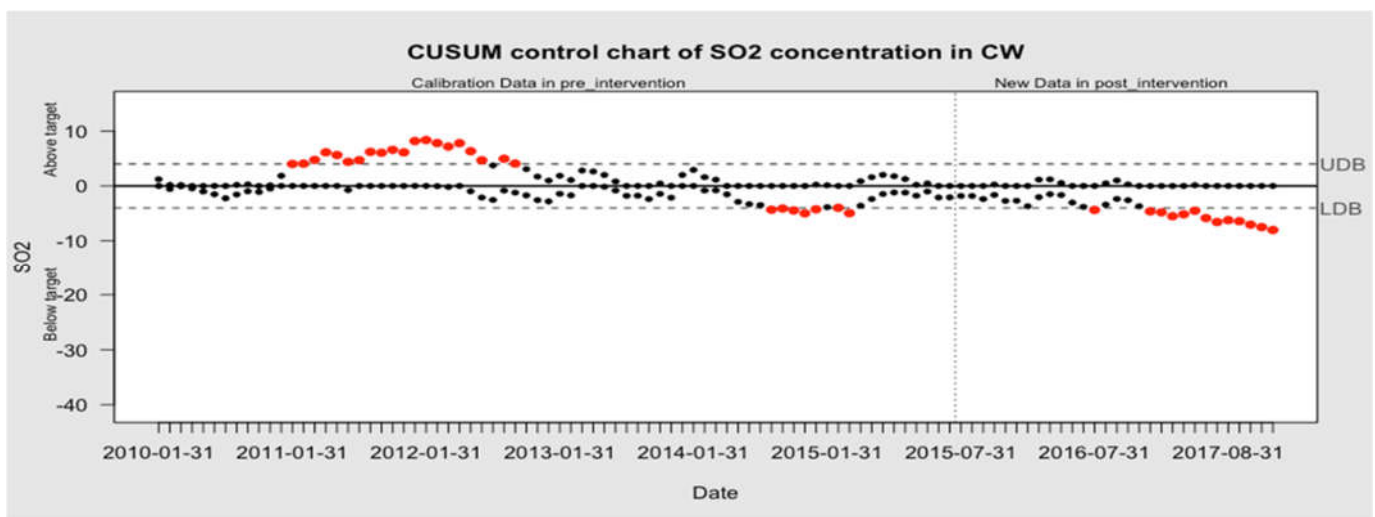




576

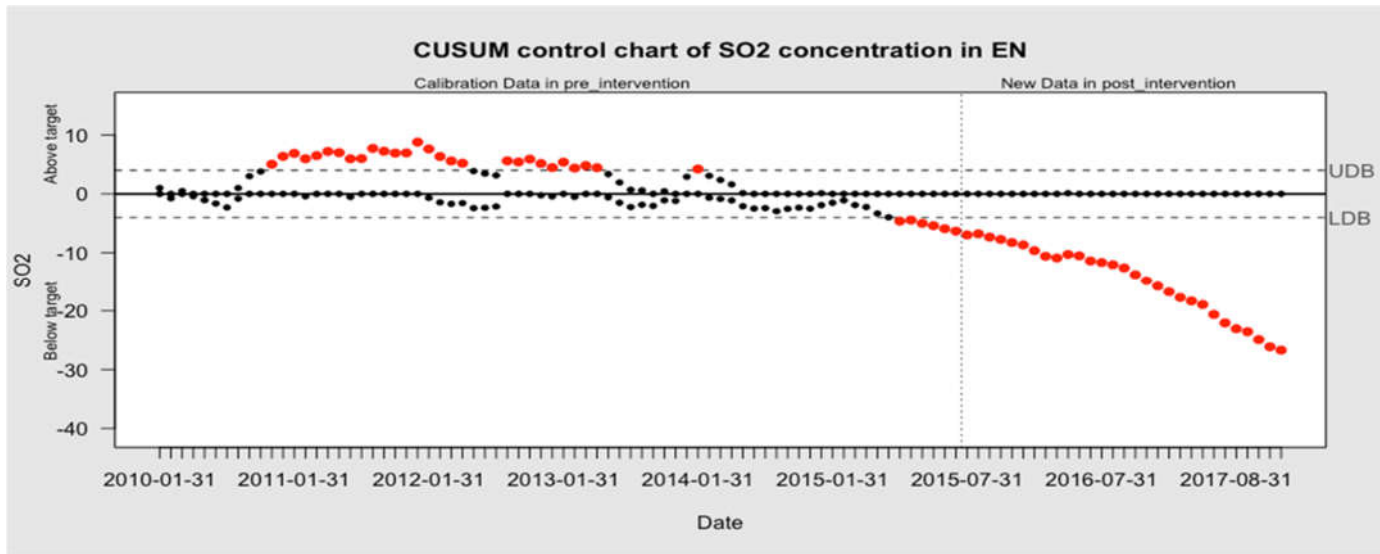
577

578

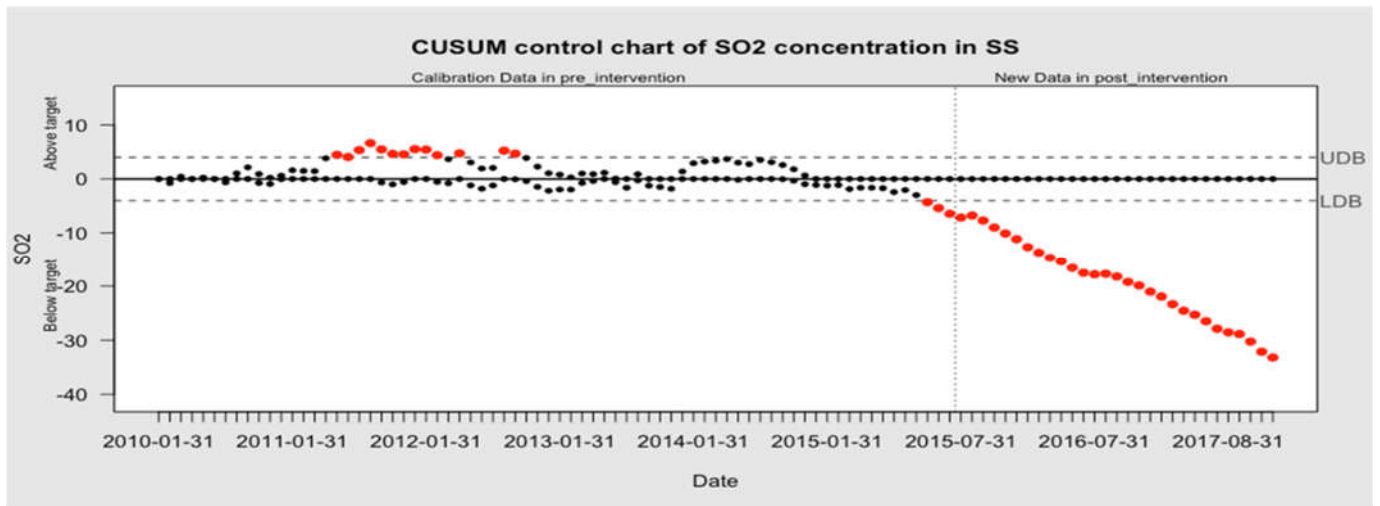


579

580



581



582

583

584

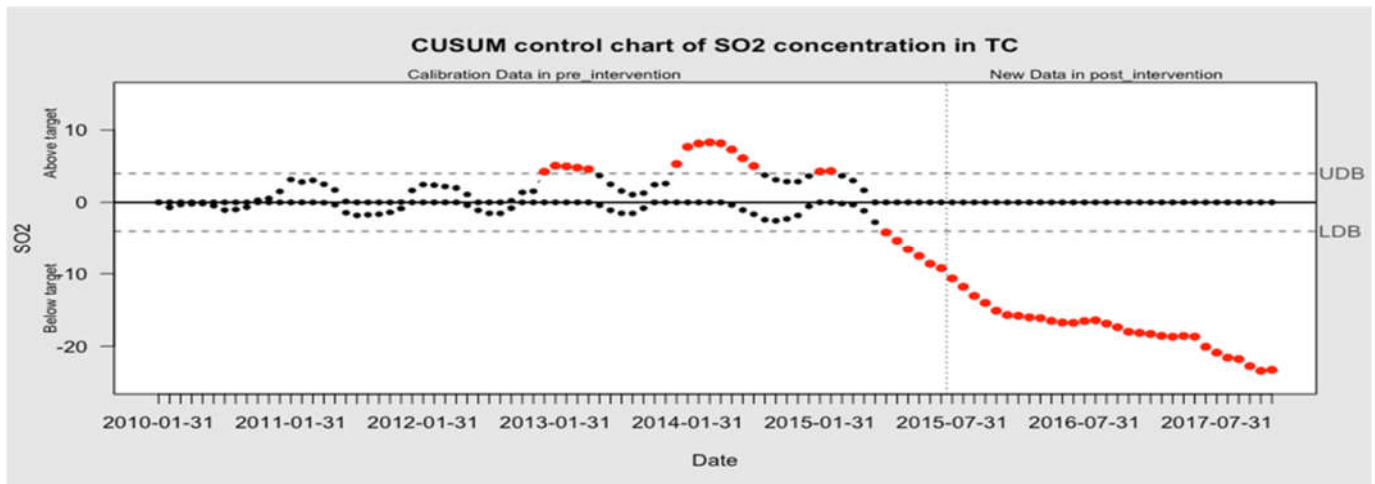
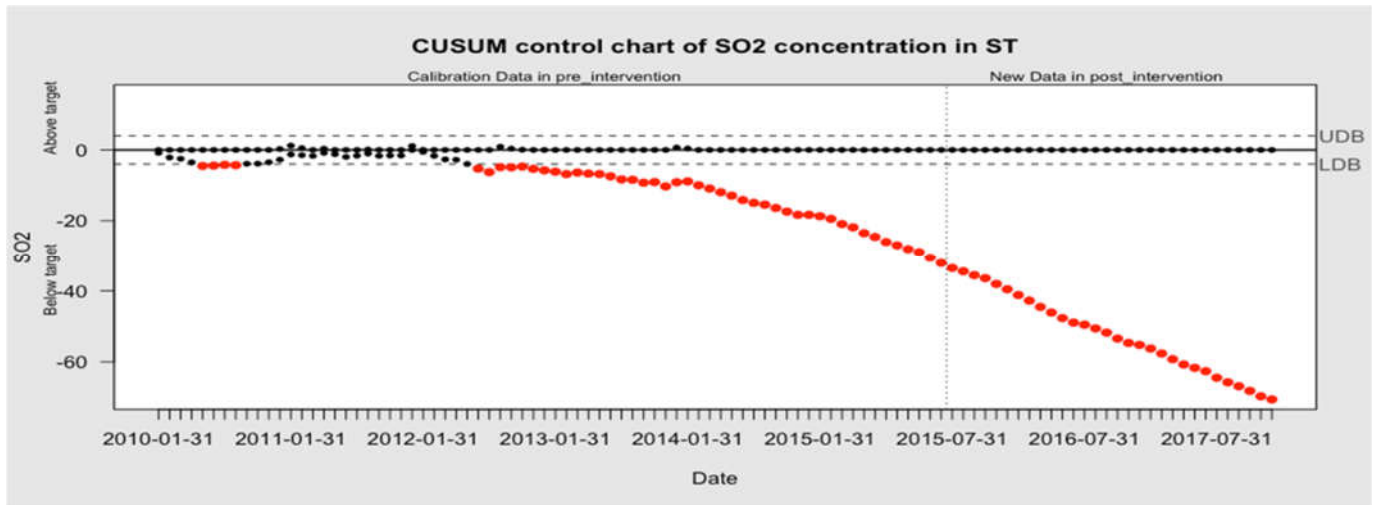


Fig.1. CUSUM charts comparing SO₂ concentrations before and after the shipping emission at all stations.